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XIAMEN
UNIVERSITY

DIGITAL LOGIC

Revision

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Arrangements

- 48 lecture hours,
- Assessment:
 - ▣ Final Exam 50%
 - ▣ Mid-term Exam 20%
 - ▣ Attendance 5%
 - ▣ Assignments 10%
 - ▣ Lab Practices 15%
- Reference books:
 - ▣ John Yarbrough, Digital logic applications and design
 - ▣ Brian Holdsworth and Clive Woods, Digital logic design.
 - ▣ 欧阳星明 《数字逻辑(第五版)》 华中科技大学出版社

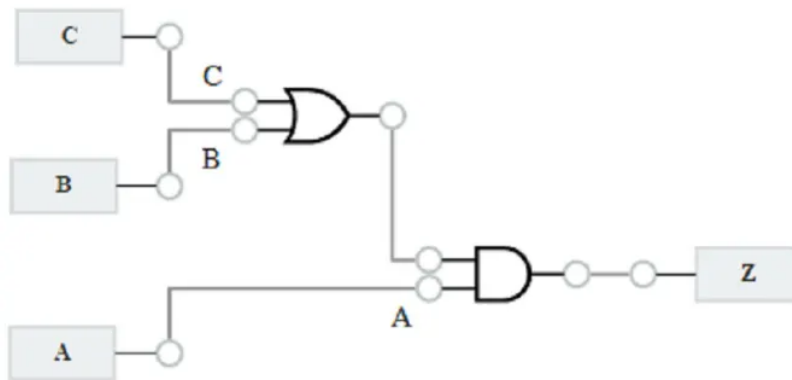
Contents

- Knowledge, basic theories, basic devices
 - Analysis and design of logic circuits based on small scale integrated circuits;
 - Understand middle - scale programmable logic devices and applications;
- **Chapter 1: Fundamental knowledge**
 - **Chapter 2: Logic Algebra**
 - **Chapter 3: Combinational Logic Circuits**
 - **Chapter 4: Flip-flop devices**
 - **Chapter 5: Synchronized Logic Circuits**
 - **Chapter 6: Asynchronized Logic Circuits**
 - ~~**Chapter 7: Middle - Scale Programmable Logic Devices**~~

Examination

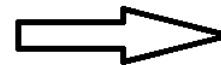
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- Four types of questions
 - ▣ Fill-in-the-blanks questions 15
 - ▣ Short answer questions 6
 - Circuit analysis questions
 - ▣ Circuit design questions 2



Diagram

Analysis



Design



$$A * (B + C)$$

Objective

How to prepare the examination



- Review knowledge key points.
- Review/redo the assignments.
- The exam content will not exceed the class content.
- The types of exam questions are the same as the those of the assignments.

Chapter 1

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- Base-r number system
 - ▣ Binary, Octal, Hex systems.
 - ▣ Conversions between each two systems, e.g., Binary to Octal, Hex and Decimal, vice versa. Octal to Hex?
 - ▣ Integers and fractions
- Signed Binary Code
 - ▣ Sign-magnitude code
 - ▣ 1s complement code
 - ▣ 2s complement code
 - ▣ Integers and fractions

Chapter 1

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- BCD codes
 - ▣ 8421
 - ▣ 2421
 - ▣ Excess 3
- Conversions from 8421 codes or 2421 codes to decimal numbers
- Grey code
 - ▣ Binary number to Gray code, vice versa
- Don't forget Subscripts

Chapter 2 Logic Algebra

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- Good understanding on Properties, Theorems, Rules, Tools
- Arithmetic Precede
- Truth Table
- Karnaugh map
- Theorems, each line uses only one theorem.
- Substitution rule
- Complement rule
- Dual rule

Chapter 2 Logic Algebra

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- Compound Logic
- Universal Gates: NAND, NOR

AND: $F = \overline{\overline{A \cdot B \cdot 1}} = \overline{\overline{A \cdot B}} = A \cdot B$

OR: $F = \overline{\overline{A \cdot 1 \cdot B \cdot 1}} = \overline{\overline{A \cdot B}} = A + B$

NOT: $F = \overline{A \cdot 1} = \overline{A}$

AND: $F = \overline{\overline{A + 0 + B + 0}} = \overline{\overline{A + B}} = A \cdot B$

OR: $F = \overline{\overline{A + B + 0}} = \overline{\overline{A + B}} = A + B$

NOT: $F = \overline{A + 0} = \overline{A}$

AND-OR-INVERT

Chapter 2 Logic Algebra

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- Common Terms
 - ▣ “AND-OR” and “OR-AND” Terms.
- Canonical Terms
 - ▣ minterms and maxterms
- Relationships between Minterm and Maxterm

$$\overline{m_i} = M_i \quad m_i = \overline{M_i}$$

- Canonical Sum of Minterms
- Canonical Product of Maxterms

<i>A</i>	<i>B</i>	<i>C</i>	Minterm	Index
0	0	0	$\overline{A} \overline{B} \overline{C}$	m_0
0	0	1	$\overline{A} \overline{B} C$	m_1
0	1	0	$\overline{A} B \overline{C}$	m_2
0	1	1	$\overline{A} B C$	m_3
1	0	0	$A \overline{B} \overline{C}$	m_4
1	0	1	$A \overline{B} C$	m_5
1	1	0	$A B \overline{C}$	m_6
1	1	1	$A B C$	m_7

<i>A</i>	<i>B</i>	<i>C</i>	Maxterm	Index
0	0	0	$A + B + C$	M_0
0	0	1	$A + B + \overline{C}$	M_1
0	1	0	$A + \overline{B} + C$	M_2
0	1	1	$A + \overline{B} + \overline{C}$	M_3
1	0	0	$\overline{A} + B + C$	M_4
1	0	1	$\overline{A} + B + \overline{C}$	M_5
1	1	0	$\overline{A} + \overline{B} + C$	M_6
1	1	1	$\overline{A} + \overline{B} + \overline{C}$	M_7

Chapter 2 Logic Algebra

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☐ ~~To obtain canonical POM~~

☐ But $F(x, y, z) = \Sigma_m(1, 3, 5, 7)$

$$F(x, y, z) = \Pi_M(0, 2, 4, 6)$$

Chapter 2 Logic Algebra

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- Simplification/Optimization of logic functions
 - ▣ Algebraic simplification
 - ▣ Karnaugh map simplification
- Plotting K-Map
- ~~Or AND~~

B \ A	0	1
	0	1
0	m_0	m_2
1	m_1	m_3

(a)

C \ AB	00	01	11	10
	0	1	1	0
0	m_0	m_2	m_6	m_4
1	m_1	m_3	m_7	m_5

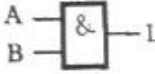
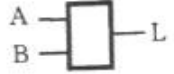
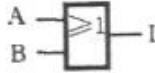
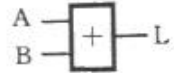
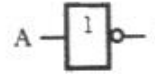
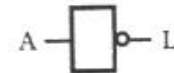
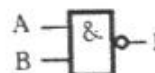
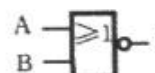
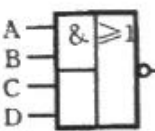
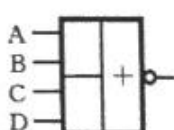
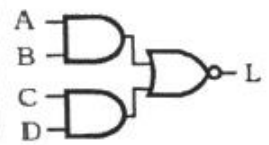
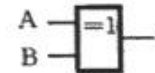
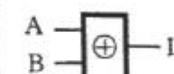
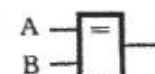
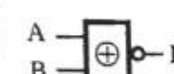
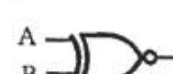
(b)

CD \ AB	00	01	11	10
	00	01	11	10
00	m_0	m_4	m_{12}	m_8
01	m_1	m_5	m_{13}	m_9
11	m_3	m_7	m_{15}	m_{11}
10	m_2	m_6	m_{14}	m_{10}

Chapter 3

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表 3.12 常用门电路符号及表达式

名称	逻辑功能	新标准符号	沿用符号	国外流行符号	逻辑表达式
与门	与运算				$L = A \cdot B$
或门	或运算				$L = A + B$
非门	非运算				$L = \bar{A}$
与非门	与非运算				$L = \overline{A \cdot B}$
或非门	或非运算				$L = \overline{A + B}$
与或非门	与或非运算				$L = \overline{AB + CD}$
异或门	异或运算				$L = A \oplus B$ $= A\bar{B} + \bar{A}B$
同或门	同或运算				$L = A \odot B$ $= \overline{A \oplus B}$ $= AB + \bar{A}\bar{B}$

Chapter 3

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☐ ~~Pinout Diagrams~~

- ☐ Determine circuit types(Combination\Asynchronous): Two Features

 - ☐ Feedback wires?

☐ Analysis of combinational logic circuits

- 1. Obtain the output function expression according to the logic circuit diagram**

- 2. Simplify output function expressions**

- 3. List the output function's truth table**

- 4. Evaluate the function**

- ☐ Don't care terms !!!

Chapter 3

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□ Combinational logic circuit design

- 1. Establish a logical description of a given problem
i.e., Truth Table**
- 2. Find the simplest expression of the logic function
i.e., K-map**
- 3. Select the logic gate type and transform the logic
function into the corresponding form
i.e., NAND? NOR?**
- 4. Plot a logic circuit diagram
pencils, rulers, erasers.**

Chapter 4

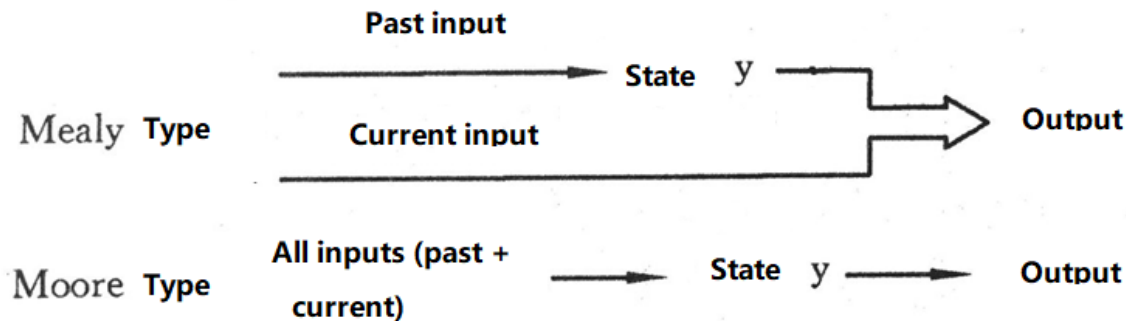
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- All flip-flops w/wo cp
- ~~Function tables/excitation tables~~
- Next State Equations & Constraint Equations
- Convert from RS to D, from Jk to T, from JK to D

Chapter 5

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- Determine circuit types:
Synchronous\Asynchronous(Pulse)
- Output, Excitation, Next state functions
- Mealy/Moore



Chapter 5

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- Basic State Table: Mealy and Moore
- Basic State Diagram: Mealy and Moore
- Conversions between State tables and State diagrams

Chapter 5

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□ Analysis

1. List the output function and the excitation function expression.

Double check here

2. List the circuit **state truth table** with the flip-flop function table.

Table Header

3. List the state table and state diagram (take a timing diagram if necessary).

Mealy/Moore

4. Summarize the logic functions of the circuit.

clock ticks:	1	2	3	4	5	6	7	8
Input x_1 :	0	0	1	1	0	1	1	0
Input x_2 :	0	1	0	1	1	1	0	0
State y :	"0"	0	0	0	1	1	1	1
Output Z :	0	1	1	0	0	1	0	1

Chapter 5

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- Design of synchronous sequential logic circuit

The general steps of the design are as follows:

1. Generate basic state diagram and basic state table;
Double check (Mealy types)
2. State reduction, to obtain the reduced state table;
Implicant Chart
3. State assignment, to obtain the reduced binary state table; 3.5 rules
4. Select flip-flops, and find the simplest expression of the excitation function and the output function; ~~NAND-NOR~~
5. Plot the logic diagram



Chapter 5

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- Strictly follow typical input /output sequence
- State Equivalence: Dependencies with all inputs
- Implicant Chart+Observation
- Quantitative Relationships between states and flip-flops
- Y-Z matrices: table header

Chapter 6 Asynchronous Logic Circuits

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- Determine circuit types:
 - ▣ Combinantion\Synchronous\Asynchronous,
 - ▣ Pulse\Level
 - ▣ Moore\Mealy
- Analysis
- ~~□ Design~~
- ~~□ Level Asynchronous~~

Chapter 6 Asynchronous Logic Circuits

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□ Analysis:

- (1) Write the output function and excitation function expression of the circuit;**
- (2) List circuit next state truth table or excitation equations;**
- (3) Generate a state table and a state diagram;**
- ~~—(4) Describe the logic function of the circuit in words (plot a timing diagram if necessary)。~~**